

Data Overview

- Rows: 202 (201 if including data only)
- Columns: 5 (Country, Region, Year, Average_Monthly_Income, Cost_of_Living):
 - □Country (object): The name of the country where the data was recorded.
 - □Region (object): The geographical region to which the country belongs
 - □Year (int): The year when the data was recorded.
 - □Average_Monthly_Income (float64): The average monthly income of individuals in USD.
 - Cost_of_Living (int): The average monthly cost of living in USD, including essentials like housing, food, and utilities.

	Country	Year	Average_Monthly_Income	Cost_of_Livin	Region
	Australia	2013	3483.92	1106.07	Oceania
	India	2019	7771.03	5422.78	Asia
	Russia	2004	6991.3	3972.36	Europe
	South Africa	2011	6628.04	6755.75	Africa
	Brazil	2015	2434.27	2656.36	South America
	Australia	2015	1781.66	4575.3	Oceania
	China	2020	5514.82	6551.39	Asia
	Japan	2006	7470.32	1076.56	Asia
0	Australia	2003	4675.72	6585.71	Oceania
1	Russia	2000	4787.1	4940.05	Europe
2	Russia	2004	2599.84	847.72	Europe
3	South Africa	2022	6271.2	2386.36	Africa
4	Brazil	2009	1902.83	5073.94	South America
5	India	2021	2927.59	844.51	Asia
6	South Africa	2004	3690.77	4242.33	Africa
7	South Africa	2003	4307.08	2682.83	Africa
8	Japan	2001	2318.07	4498.04	Asia
9	Canada	2019	1361.28	701.9	North America
0	Brazil	2009	5079.65	6152.14	South America
1	Germany	2018	2664.73	6825.03	Europe
2	South Africa	2000	4859.29	6794.59	Africa
3	Mexico	2023	1657.72	5347.7	North America

- Context: The dataset compares countries' average monthly income and cost of living across regions and years — useful for analyzing affordability or income-to-cost ratios globally.

Data Cleaning

- **Missing values:**

- Average_Monthly_Income: 2 missing

2007	1822.9	6
2015	4237.76	6
2012		4
2000	7361.34	1
2021	3217.95	4
2019	4854.41	8
2016	5241.98	6
2006	598.21	2
2012	5476.53	3
2003	1835.27	6
2003	7708.03	5
2005	1614.97	5
2018	3609.68	3
2011	1140.12	2
2021	7976.56	2
2006	4266.46	6
2009	4965.39	6
2018	1003.07	6
2006		4
2002	2074.29	3
2012	7235.41	5

* This can be handled by using the median income of that region or global median.

15	4237.76
12	4270.4
00	7361.34
21	3217.95
19	4854.41
16	5241.98
06	598.21
12	5476.53
03	1835.27
03	7708.03
05	1614.97
18	3609.68
11	1140.12
21	7976.56
06	4266.46
09	4965.39
18	1003.07
06	4270.4
02	2074.29
12	7235.41

■ Region: 2 missing

7.7	North America	
.57	North America	
.54	Europe	
.27		
.02	South America	
.83	North America	
.05	North America	
.69	Asia	
.75	North America	
.96	North America	
.94	Europe	
.02	North America	
2.8	Europe	
.15	Europe	
.69	Asia	
.92	Asia	
.74		
3.3	North America	

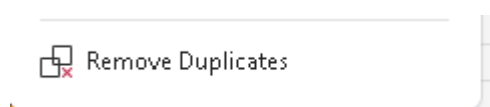
*This can be handled by putting region based on the country of that data.

● Duplicate data:

■ There are 2 duplicates:

28	United State	2023	1508.11	2535.83	North America
29	Mexico	2015	975.31	3625.05	North America
30	China	2022	7924.7	5484.69	Asia
31	Canada	2001	2917.65	4909.75	North America
32	Mexico	2016	6574.06	3342.96	North America
33	France	2019	2409.8	2205.94	Europe
34	United State	2023	5611.27	6981.02	North America
35	Russia	2011	6201.71	3212.8	Europe
36	Russia	2017	4967.29	3379.15	Europe
37	China	2022	7924.7	5484.69	Asia
38	China	2002	4036.82	1479.92	Asia
39	Mexico	2000	3588.81	5645.74	North America
40	Mexico	2000	3116.51	4978.3	North America
41	Japan	2018	7471.47	1857.08	Asia
42	Mexico	2010	6729.65	943.71	North America
43	Australia	2004	7737.7	4891.3	Oceania
44	India	2011	1432.23	4719.77	Asia
45	Mexico	2010	6729.65	943.71	North America
46	France	2002	5981.51	2203.51	Europe
47	Japan	2000	7537.55	6675.7	Asia
48	Brazil	2000	1850.05	1206.08	South America

● We will handle this using the remove duplicate feature:



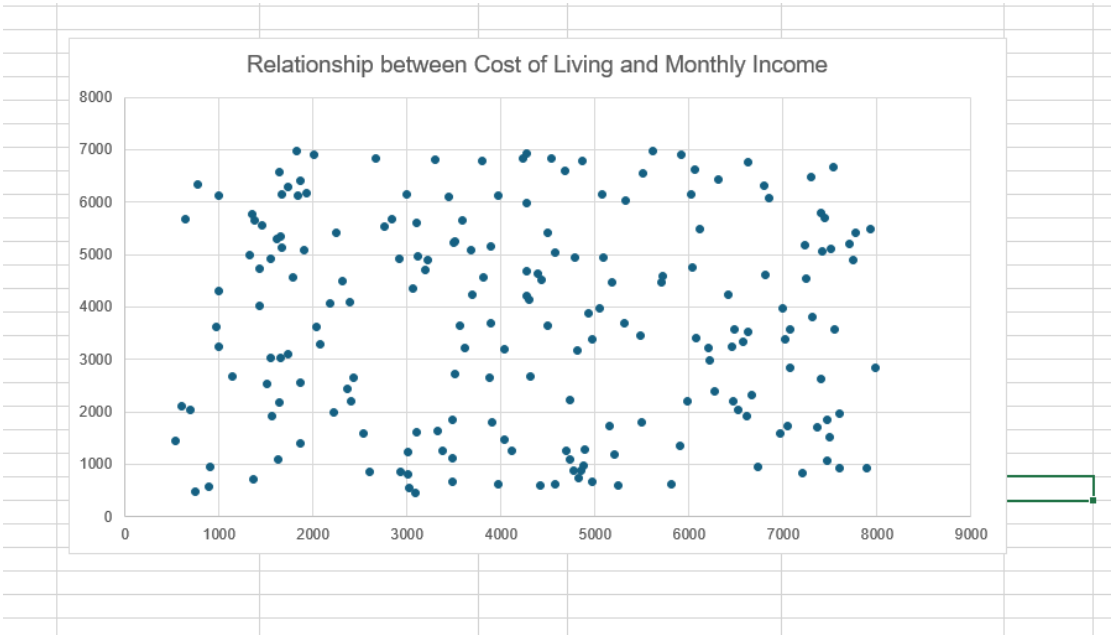
Descriptive Statistics

Average Monthly Income		Cost of Living	
Mean	4244.193216	Mean	3705.127286
Standard Error	150.0444918	Standard Error	140.515613
Median	4266.46	Median	3688.09
Mode	4266.46	Mode	#N/A
Standard Deviation	2116.638031	Standard Deviation	1982.216654
Sample Variance	4480156.553	Sample Variance	3929182.864
Kurtosis	-1.166687671	Kurtosis	-1.272006829
Skewness	0.055307916	Skewness	-0.04103302
Range	7441.82	Range	6516.53
Minimum	534.74	Minimum	464.49
Maximum	7976.56	Maximum	6981.02
Sum	844594.45	Sum	737320.33
Count	199	Count	199

The average monthly income is \$4,244, compared to an average cost of living of \$3,705. Both figures exhibit high standard deviations (Income: \$2,116.64; Cost: \$1,982.22), reflecting considerable variation across countries. This indicates notable economic inequality, where some nations maintain a comfortable gap between income and living costs, while others face challenges due to lower incomes or higher expenses.

Insights

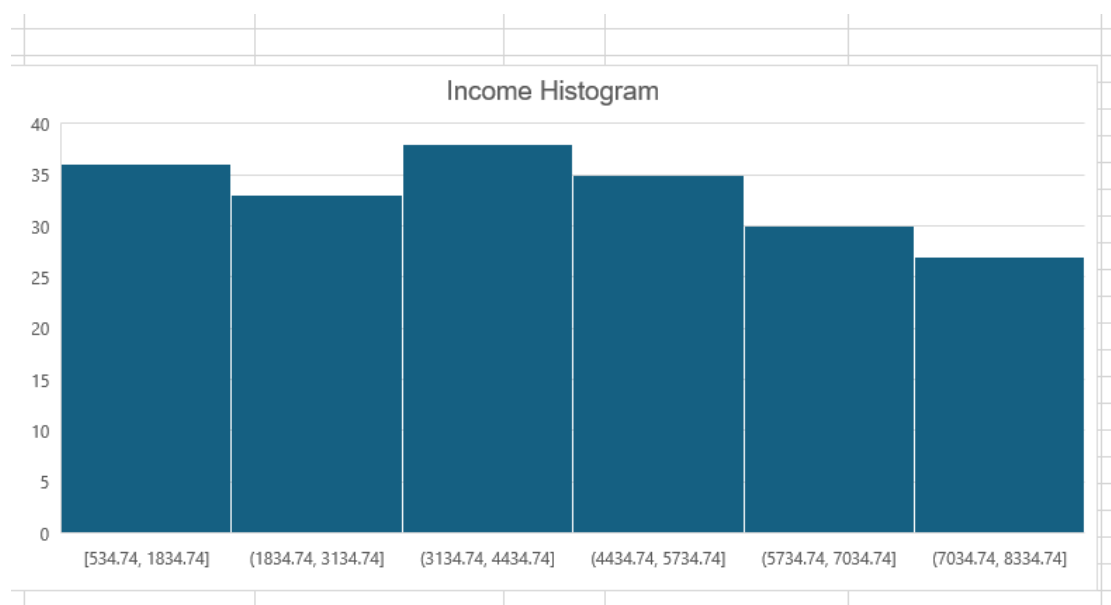
1. Relationship between Cost of Living and Income



The scatter plot shows that there is little-to-no relationship between Average Monthly Income and Cost of Living. Countries with high incomes do not always have high living costs, and some low-income countries still experience relatively high expenses. This suggests that other factors — such as economic structure, regional prices, or currency strength — may influence living costs more than income alone.

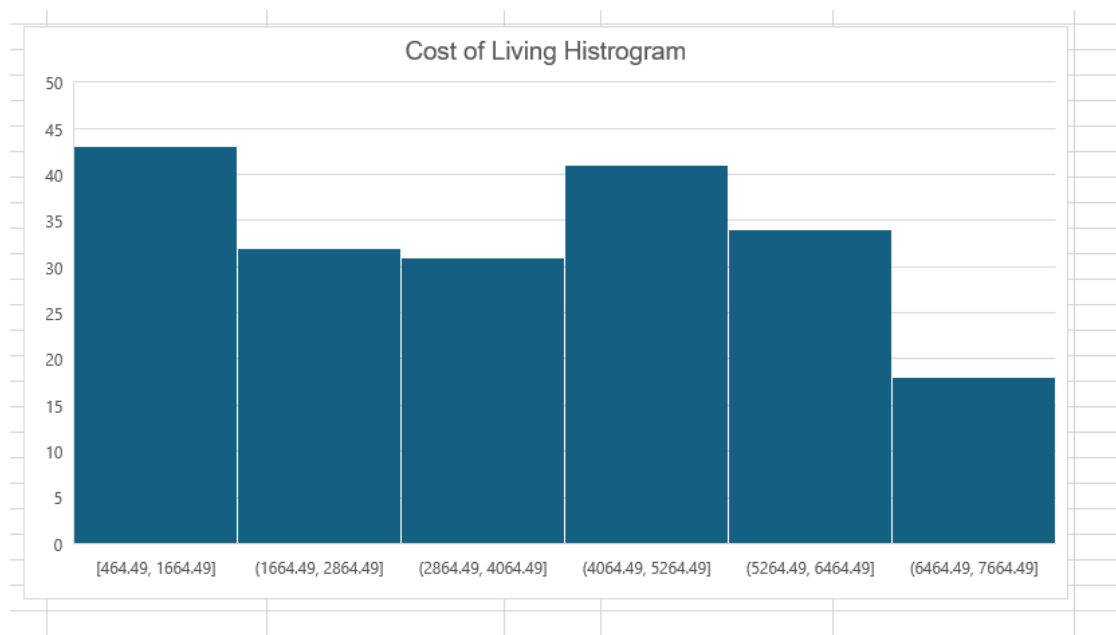
2. Histograms for Income and Cost

2.1. Income



The histogram shows that most countries have average monthly incomes between \$3,000 and \$4,500, with fewer at the very high or low ends. This indicates a moderate concentration of income levels globally, suggesting limited extreme disparities.

2.2. Cost



Most countries have a cost of living between \$4,000 and \$5,000, with fewer outliers on either side. This suggests that while living expenses vary, they remain relatively balanced across most nations.